

Direct Generation of a Somos–4 Sequence from an Algebraic Generating Function

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Abstract

We record a five-parameter quadratic algebraic generating function whose coefficient sequence has prescribed initial Hankel determinants (λ, m, r, η) and whose Hankel transform belongs to the Somos–4 family $A(1, \tau)$. The Hankel determinants and their parameters are introduced first, before the explicit algebraic equation. We then describe the two square-root branches, their coalescence at the origin, a desingularized coefficient recurrence, the induced shift of the Hankel parameters, the Hankel-transform identity used in the construction, and the nonuniqueness of the coefficient sequence underlying a given Hankel transform.

1 Hankel determinants and Somos parameters

Let

$$G(z) = \sum_{n \geq 0} g_n z^n$$

be a formal power series, and for $n \geq 1$ define its ordinary Hankel determinants by

$$H_n = \det(g_{i+j})_{0 \leq i, j \leq n-1}, \quad H_0 = 1. \quad (1)$$

The five parameters $\lambda, m, r, \eta, \tau$ have the following roles:

$$H_1 = \lambda, \quad H_2 = m, \quad H_3 = r, \quad H_4 = \eta, \quad (2)$$

and the Hankel sequence satisfies the Somos–4 recurrence

$$H_{n+4}H_n = H_{n+3}H_{n+1} + \tau H_{n+2}^2. \quad (3)$$

Thus τ is the second coefficient of the Somos family $A(1, \tau)$, whereas λ, m, r , and η select the initial Hankel determinants. The algebraic generating function producing these determinants is given in the next section.

2 The algebraic generating function

Define the polynomials $A(z), B(z), C(z)$ by

$$A(z) = \lambda \eta m r^2 (\tau - \eta) z^3 \quad (4)$$

$$+ r \left(-(\lambda^2 \eta m + m^2(m + r^2))\tau + (2m - 1)\lambda^2 \eta^2 + \eta m^2(m + r^2) - \lambda r(m + r^2) \right) z^2 \quad (5)$$

$$+ \lambda \left(m^2 r^2 \tau - (m - 1)\lambda^2 \eta^2 - m(m^2 + 2mr^2 - m - r^2)\eta + \lambda r^3 \right) z \quad (6)$$

$$+ \lambda^2 \eta m(m - 1)r, \quad (7)$$

$$B(z) = -\lambda^2 \eta m r^2 \tau z^3 \quad (8)$$

$$+ r \left((2\lambda^3 \eta m + \lambda m^2(2m + r^2))\tau - (2m - 1)\lambda^3 \eta^2 - \lambda \eta m^3 + \lambda^2 r(2m + r^2) \right) z^2 \quad (9)$$

$$+ \lambda^2 \left(-2m^2 r^2 \tau + 2(m - 1)\lambda^2 \eta^2 + m(2m^2 + 2mr^2 - 2m - r^2)\eta - 2\lambda r^3 \right) z \quad (10)$$

$$- 2\lambda^3 \eta m(m - 1)r, \quad (11)$$

$$C(z) = -\lambda^2 m r \left(\tau(\lambda^2 \eta + m^2) + \lambda r \right) z^2 \quad (12)$$

$$+ \left(\lambda^3 m^2 r^2 \tau + \lambda^4 r^3 + \eta(m - 1)(\eta(-15\lambda^4 + 85\lambda^3 - 225\lambda^2 + 274\lambda - 120) \right. \quad (13)$$

$$\left. - \lambda^3 m^2 - 3(\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)(\lambda - 5) \right) z \quad (14)$$

$$+ \lambda^4 \eta m(m - 1)r. \quad (15)$$

The generating function is the formal-power-series branch of

$$A(z)G(z)^2 + B(z)G(z) + C(z) = 0 \quad (16)$$

selected by the initial conditions implicit in (2), beginning with $g_0 = \lambda$ and $g_1 = m$.

Remark 1 (The apparently mysterious integer coefficients). The polynomial occurring in (15) is simply

$$-15\lambda^4 + 85\lambda^3 - 225\lambda^2 + 274\lambda - 120 = \frac{6!}{\lambda} \binom{\lambda}{6} - \lambda^5 \quad (17)$$

$$= (\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)(\lambda - 5) - \lambda^5. \quad (18)$$

Thus the coefficients $-15, 85, -225, 274, -120$ arise by removing the leading term λ^5 from the falling factorial $(\lambda - 1) \cdots (\lambda - 5)$.

3 The two algebraic branches

With

$$D(z) = B(z)^2 - 4A(z)C(z),$$

the two algebraic branches may be represented by

$$G(z) = \frac{-B(z) \pm \sqrt{D(z)}}{2A(z)}. \quad (19)$$

When the constant term of A vanishes, the rationalized expressions

$$\frac{-2C(z)}{B(z) \mp \sqrt{D(z)}}$$

can be used instead. The sign in (19) cannot be fixed uniformly over the full parameter space.

Write

$$P(z, X) = A(z)X^2 + B(z)X + C(z).$$

The constant coefficients of (7)–(15) are

$$A_0 = \lambda^2 \eta m(m-1)r, \quad B_0 = -2\lambda^3 \eta m(m-1)r, \quad C_0 = \lambda^4 \eta m(m-1)r.$$

Consequently,

$$P(0, X) = A_0(X - \lambda)^2. \quad (20)$$

Both algebraic branches therefore coalesce at $X = \lambda$ when $z = 0$, and

$$\partial_X P(0, \lambda) = 2A_0\lambda + B_0 = 0. \quad (21)$$

The simple-root form of the formal implicit-function theorem is not applicable at the origin. The desired branch is instead selected by expanding the square root and requiring the prescribed coefficients, beginning with $g_0 = \lambda$ and $g_1 = m$.

Set

$$G(z) = \lambda + zF(z), \quad F(z) = \sum_{q \geq 0} f_q z^q, \quad f_q = g_{q+1}. \quad (22)$$

For

$$D_j = 2\lambda A_j + B_j, \quad E_j = \lambda^2 A_j + \lambda B_j + C_j,$$

where $C_3 = 0$, substitution into $P(z, G(z)) = 0$ gives

$$P(z, \lambda + zF) = \sum_{j=0}^3 E_j z^j + z \left(\sum_{j=0}^3 D_j z^j \right) F + z^2 \left(\sum_{j=0}^3 A_j z^j \right) F^2.$$

Equation (20) gives $E_0 = D_0 = 0$. An ordinary power-series branch additionally requires

$$E_1 = 0. \quad (23)$$

The next coefficient equation is

$$A_0 f_0^2 + D_1 f_0 + E_2 = 0. \quad (24)$$

Since $f_0 = g_1 = m$, the prescribed value of m selects one of the two branches of the desingularized quadratic.

4 Hankel-level shift of the parameters

To display the shift independently of the letters used above, let

$$G_{0;\ell,m,r,n,t}(x) = \sum_{q \geq 0} g_q x^q$$

denote the selected formal-power-series solution of (16) after the substitutions

$$\lambda = \ell, \quad \eta = n, \quad \tau = t.$$

Its Hankel determinants satisfy

$$(H_1, H_2, H_3, H_4) = (\ell, m, r, n), \quad H_{q+4}H_q = H_{q+3}H_{q+1} + tH_{q+2}^2. \quad (25)$$

The first Somos step gives

$$H_5 = \frac{H_2 H_4 + t H_3^2}{H_1} = \frac{mn + tr^2}{\ell}. \quad (26)$$

Hence shifting the Hankel sequence induces the parameter map

$$(\ell, m, r, n, t) \mapsto \left(m, r, n, \frac{mn + tr^2}{\ell}, t \right). \quad (27)$$

The point of this shift in the present construction is the following Hankel-transform identity. Remove the first coefficient of $G_{0;\ell,m,r,n,t}$ and shift the remaining series to the origin:

$$\widehat{G}(x) = \frac{G_{0;\ell,m,r,n,t}(x) - g_0}{x} = \sum_{q \geq 0} g_{q+1} x^q. \quad (28)$$

Also form the generating function with shifted parameters,

$$\widetilde{G}(x) = G_{0;m,r,n,(mn+tr^2)/\ell,t}(x) = \sum_{q \geq 0} \widetilde{g}_q x^q. \quad (29)$$

Although the two coefficient sequences are generally different, their Hankel transforms are identical:

$$\det(g_{i+j+1})_{0 \leq i,j \leq q-1} = \det(\widetilde{g}_{i+j})_{0 \leq i,j \leq q-1} \quad (q \geq 1). \quad (30)$$

Both sides therefore begin with

$$m, \quad r, \quad n, \quad \frac{mn + tr^2}{\ell}, \dots$$

and continue according to the same Somos–4 recurrence with parameter t . This Hankel-level identity, rather than equality of the generating functions, is the shift property used in the design of the algebraic generating function. In general,

$$\widehat{G}(x) \neq \widetilde{G}(x),$$

which is consistent with the fact that different coefficient sequences can have the same Hankel transform.

5 A recurrence for the coefficients

Using the notation of (22), division of the expanded quadratic equation by z^2 yields

$$A(z)F(z)^2 + (D_1 + D_2 z + D_3 z^2)F(z) + E_2 + E_3 z = 0. \quad (31)$$

The branch condition is

$$A_0 m^2 + D_1 m + E_2 = 0, \quad (32)$$

and the branch is simple precisely when

$$2A_0 m + D_1 \neq 0. \quad (33)$$

Equivalently,

$$(2A_0 m + D_1)^2 = D_1^2 - 4A_0 E_2. \quad (34)$$

Define

$$T_q = \sum_{k=0}^q f_k f_{q-k}, \quad T_q = 0 \quad (q < 0). \quad (35)$$

Coefficient extraction from (31) gives

$$\sum_{j=0}^3 A_j T_{q-j} + D_1 f_q + D_2 f_{q-1} + D_3 f_{q-2} + E_{q+2} = 0, \quad (36)$$

where $f_q = 0$ for $q < 0$ and $E_j = 0$ for $j > 3$. For $q \geq 1$,

$$T_q = 2m f_q + \sum_{k=1}^{q-1} f_k f_{q-k}. \quad (37)$$

Under condition (33), the coefficients are therefore generated recursively by

$$f_q = - \frac{A_0 \sum_{k=1}^{q-1} f_k f_{q-k} + \sum_{j=1}^3 A_j T_{q-j} + D_2 f_{q-1} + D_3 f_{q-2} + E_{q+2}}{2A_0 m + D_1}. \quad (38)$$

This quadratic convolution recurrence follows directly from the algebraic generating-function equation. On the exceptional locus $2A_0 m + D_1 = 0$, the two desingularized branches coalesce at the origin and the next coefficient is not determined by the first-order coefficient equation; higher-order compatibility or a Puiseux expansion may then be required.

Because F is algebraic of degree at most two, it is D-finite. Consequently, the same coefficient sequence also satisfies a fixed-width linear recurrence with polynomial coefficients in the index.

6 Nonuniqueness of the coefficient sequence

The Hankel transform does not in general determine the coefficient sequence. The five-parameter algebraic generating function should therefore be viewed as a representative of a larger equivalence class, rather than as the unique generating function of a Somos orbit. For example, the binomial transform

$$g'_q = \sum_{k=0}^q \binom{q}{k} g_k \quad (39)$$

has generating function

$$G'(z) = \frac{1}{1-z} G\left(\frac{z}{1-z}\right)$$

and preserves the Hankel determinants. Scalar and exponential rescalings of the coefficients produce explicit normalization factors in the determinants, and substitutions in the generating function give further related coefficient sequences. Equality of Hankel transforms must therefore not be replaced by equality of generating functions without an independent argument.

No classification of these equivalent coefficient sequences is attempted here. The inverse problem of describing all coefficient sequences realizing a given Hankel sequence remains open.